Unit 4, Lesson 2: Finding the Area of Trapezoids, Parallelograms, and Rhombi

Benchmark

MA.7.GR.1.1 Apply formulas to find the areas of trapezoids, parallelograms, and rhombi.

Learning Target

- ☐ I can find the area of trapezoids, parallelograms, and rhombi.

4.2.1 Warm-Up: A Day in the Life

Betsy McIver, a mechanical engineer, works in the Walt Disney Imagineering area and designs systems for the interactive attractions at the theme park. She mentions that her job can include creative left-brain thinking, as well as technical right-brain thinking. She says that to be successful at her job, it is important to understand math, design, and theatre. She also mentions that many disciplines, such as designers and architects, all work together to make everything possible in the entertainment engineering industry.

1. Describe a ride you saw that looked exciting and entertaining.

4.2.2 Guided Practice: Finding the Area of Parallelograms, Trapezoids, and Rhombi

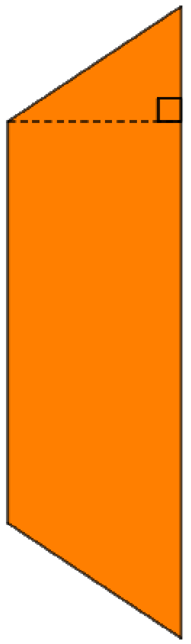
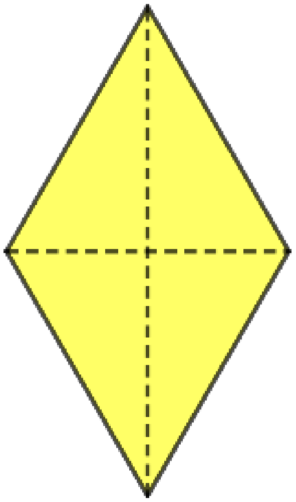
1. Complete the table to match each type of quadrilateral with the formula to find its area.

$A = \frac{1}{2}h(b_1 + b_2)$ $A = \frac{1}{2}d_1 d_2$ $A = bh$

Quadrilateral	Formula for the Area
parallelogram	
trapezoid	
rhombus	

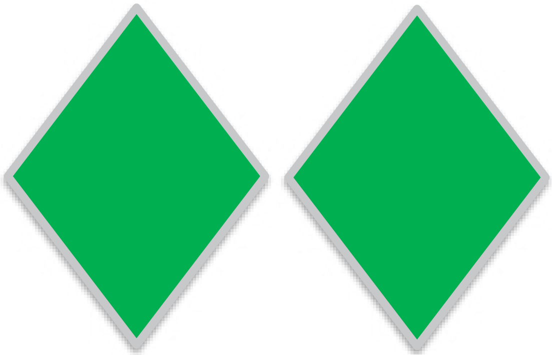
2. The dimensions of three quadrilaterals are given in the table. Label each figure shown with the given dimensions in centimeters (cm). Then, calculate the area of each figure using the appropriate formula. Show your work.

Figure	Dimensions	Area
Parallelogram 	$b = 22$ cm $h = 8$ cm	

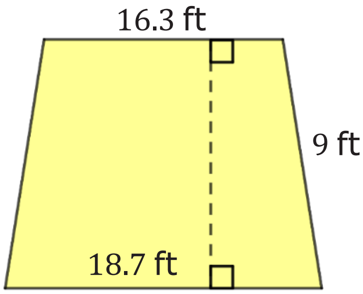
Figure	Dimensions	Area
Trapezoid 	$b_1 = \frac{7}{12}$ cm $b_2 = \frac{11}{12}$ cm $h = \frac{1}{4}$ cm	
Rhombus 	$d_1 = 6.5$ cm $d_2 = 12.25$ cm	

4.2.3 Your Turn!

1. Alina is making rhombus-shaped earrings with green stones. The measure of one diagonal of each earring is $1\frac{3}{8}$ centimeters (cm), and the length of the other diagonal measures $1\frac{4}{9}$ cm. What is the area, in square centimeters (cm^2), of two earrings?

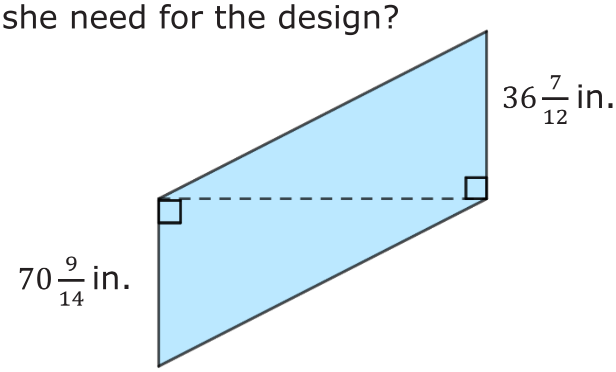


2. One wall of Kel's apartment is shaped like a trapezoid with one base 18.7 feet (ft), the other base 16.3 ft, and a height of 9 ft.



Kel plans to paint this wall blue and needs to know how much paint to buy. What is the area of the wall Kel plans to paint?

3. Mika, a designer, is creating a geometric design in a tile floor. The design will be in the shape of a parallelogram with $36\frac{7}{12}$ inch (in.) bases and a height of $70\frac{9}{14}$ in. How many square inches of tile will she need for the design?



4. The Smart family's horse pasture is in the shape of a trapezoid. One of the parallel sides is 168.2 yards (yd) long and the other is 135.6 yd long. The distance between these sides is 93.4 yd.

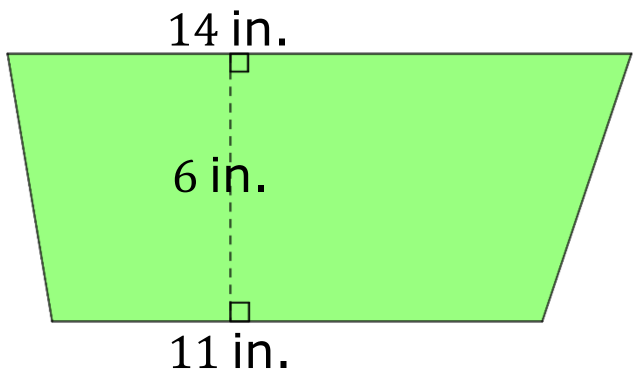
What is the area of the horse pasture?

5. The Dockland Building in Hamburg, Germany is shaped like a parallelogram.

If the base of the building is 86 meters (m) and its height is 47 m, what is the area of the face of the building shown in the image?

4.2.4 Wrap-Up: Ticket Out the Door

1. Don's work for finding the area of the trapezoid is shown below.



Don's Work
$A = \frac{1}{2}h(b_1 + b_2)$
$A = \frac{1}{2}(14)(11 + 6)$
$A = \frac{1}{2}(14)(17)$
$A = \frac{1}{2} \cdot 238$
$A = 119 \text{ in}^2$

a. Write Don a note explaining what he did correctly and where he made an error.

b. What is the area of the trapezoid?